

Artificial Intelligence

Expert Systems



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What is an Expert System?

- Human experts have
  - a considerable knowledge about their areas of expertise
  - Can learn from their experience
  - Can do reasoning
  - Can explain the solution
  - Can restructure knowledge
  - Can determine relevance



What is an Expert System?

- An Expert System (ES) is software that *attempts to reproduce the performance of one or more human experts*, typically in a specific problem domain
- a kind of software that simulates the problem-solving behavior of a human expert of given domain.
- ES employs human knowledge represented in a computer to *solve problems that ordinarily require human expertise*.
- ES imitate the expert's reasoning processes to solve specific problems
- An expert system compared with traditional computer :

Inference engine +Knowledge = Expert system

(Algorithm + Data structures= Program in traditional computer)



Why should we Use Expert System?

- Expert Systems:
  - Capture and preserve irreplaceable human expertise
  - Provide expertise needed at a number of locations at the same time or in a hostile environment that is dangerous to human health
  - Provide unemotional objective solutions faster than human experts
  - Provide expertise that is expensive or rare
  - Share human expertise with a large number of people



Early ES Systems

- First expert system, called DENDRAL, was developed in the early 70's at Stanford University.
- DENDRAL
  - the first knowledge intensive system
  - Expert system used for chemical analysis to predict molecular structure
  - determining 3D structures of complex chemical compounds
- MYCIN
  - Diagnose infectious diseases such as bacteremia and meningitis.
  - Recommend antibiotics.
  - Dosage adjusted for patient's body weight.
  - Name derived from antibiotics (suffix – "mycin").



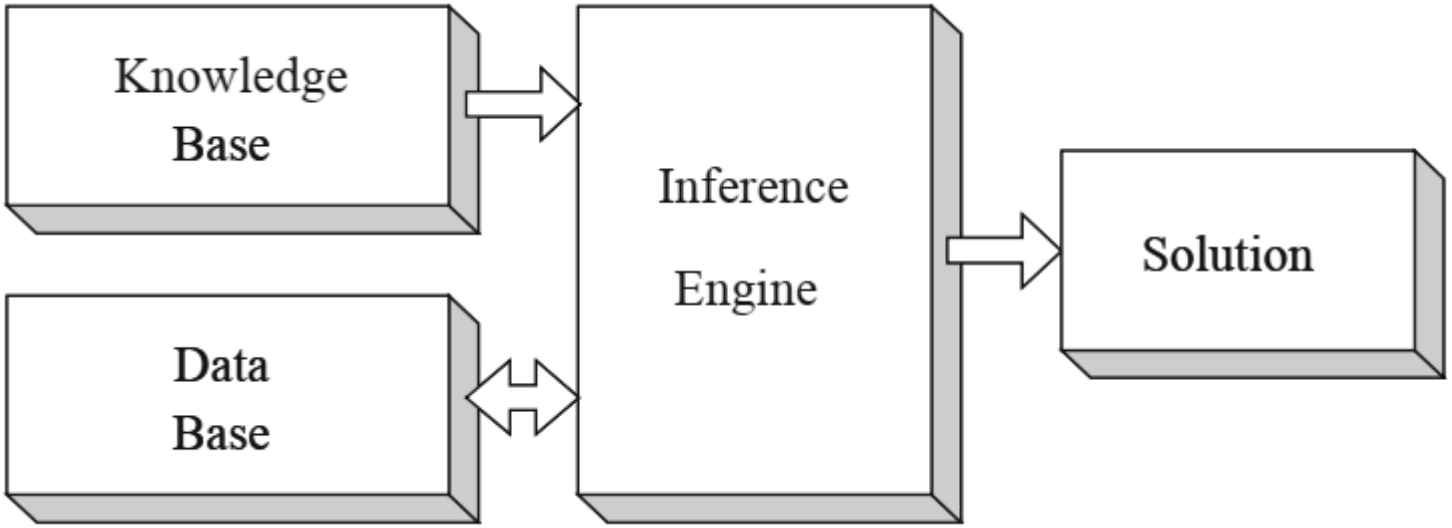
ES Systems

- SHINE: designed by NASA for monitoring, analyzing, and diagnosing real-time and non-real-time systems
- Stock Market Prediction
- Apple's SIRI, a dialog system
- insurance company Blue Cross's automated insurance claim processing system
- Some basic variants of expert systems
  - *a rule-based expert system*
  - *fuzzy expert system*
  - *neural expert system*
  - *neuro-fuzzy expert system*



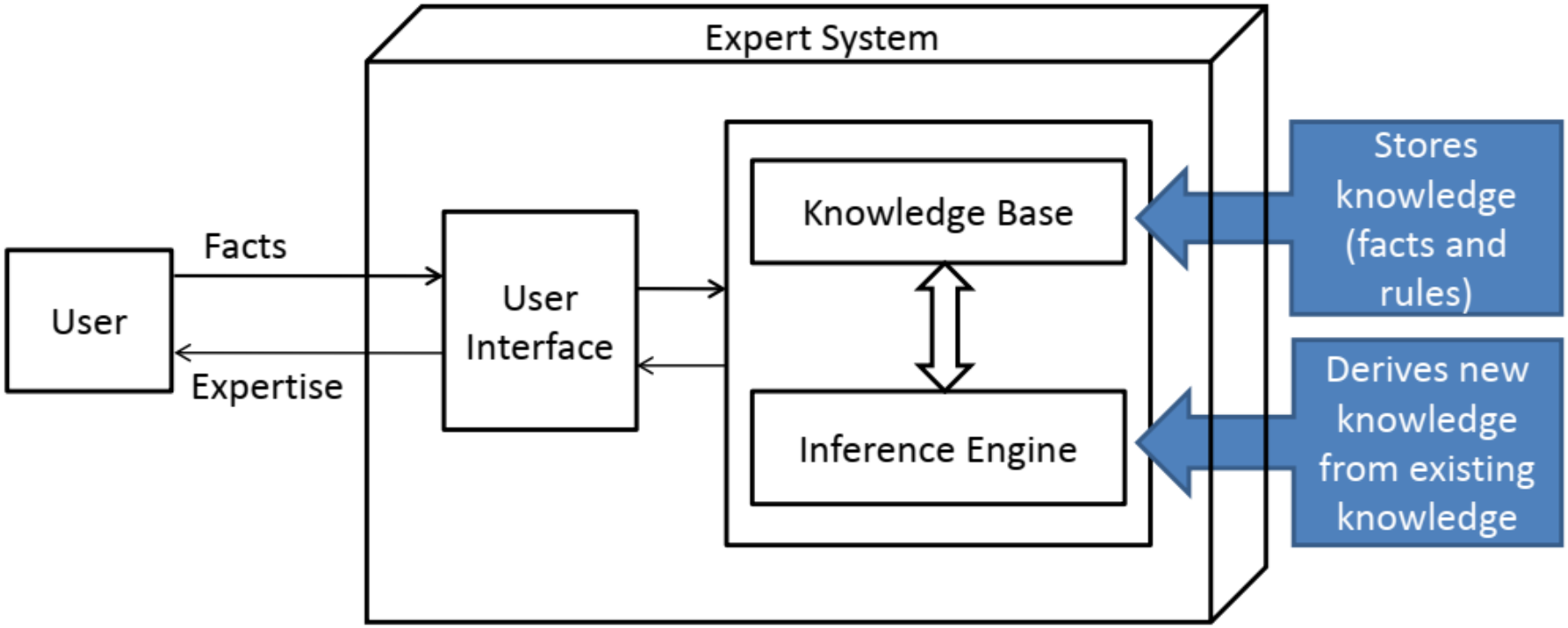
ATM machine is an Expert System???

General Structure of an ES



- Knowledge base—containing the knowledge about the system, its functioning, rules of problem solving, etc,
- Data base—including the facts, which generally describe the domain and the state of the problem to be solved
- Inference Engine—with the reasoning principles and conflict resolution strategies

Components of an Expert System



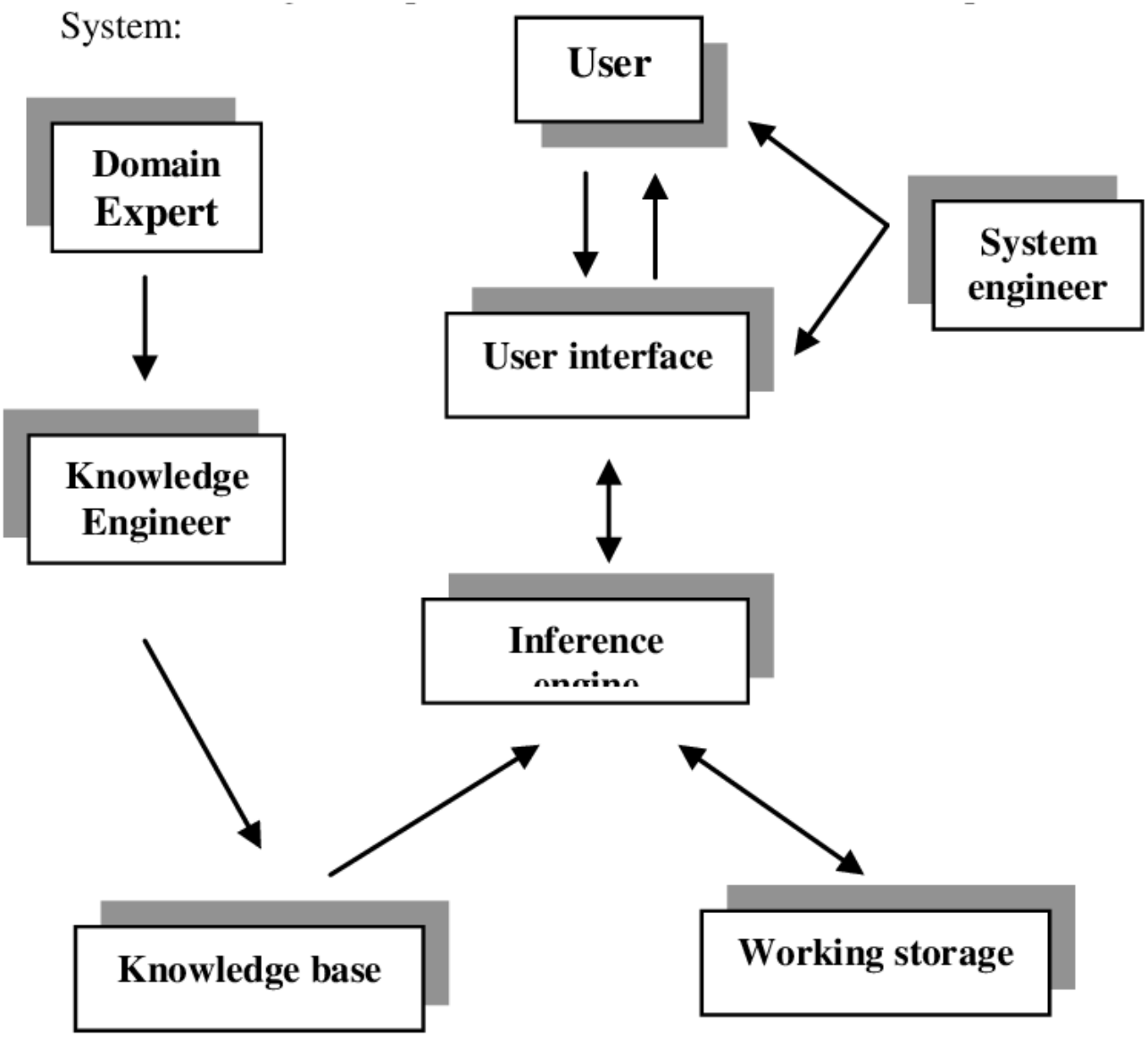
a knowledge acquisition and learning module

Designing of an Expert System

- Human expert (can solve problems; we desire to solve the problems without him/her),
- Knowledge engineer (can communicate with human expert to obtain and model the knowledge that we need in the system),
- Programmer (builds and maintains all the necessary computer programs),
- User (wants to use expertise to solve problems).



Designing of an Expert System



Explanation in Reasoning

- ES answers two questions:
  - WHY? means “why are you asking for a particular information”.
    - ES returns the current rule that is being fired.
  - HOW? means “how did you reach the conclusion”.
    - ES returns the sequence of rules that are fired to reach a conclusion.



Conventional System vs. Expert System

Conventional System	Expert System
Knowledge and processing are combined in one unit.	Knowledge database and the processing mechanism are two separate components.
The program does not make errors (Unless error in programming).	The Expert System may make a mistake.
The system is operational only when fully developed.	The expert system is optimized on an ongoing basis and can be launched with a small number of rules.
Step by step execution according to fixed algorithms is required.	Execution is done logically & heuristically.
It needs full information.	It can be functional with sufficient or insufficient information.

Human Expert Vs Expert System

Human Expert	Expert System
Resources of human expert are costly.	Expert system's resources are inexpensive
Harder to move any data and files	Easy to transfer
It is unpredictable	But, it is consistent
Harder to keep document	East to keep document
It leads to Perishable nature.	But, it leads to permanently nature.



Applications

- Medical diagnosis
- Agriculture
- Education
- Environment
- Accounting - Loan analysis
- Coding
- Games
- Planning and scheduling - Airline scheduling & cargo schedules



Limitations of the Expert System

- Limited to relatively narrow problems
- Cannot readily deal with “mixed” knowledge
- Errors in the knowledge base can lead to wrong decision
- Cannot refine own knowledge base or learn from experience
- Lack common sense
- Cannot make creative responses as human expert
- The maintenance cost of an expert system is too expensive



The End…

